METRIC

MIL-PRF-62625A(AT)
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PERFORMANCE SPECIFICATION

VALVE, FUEL-DRAIN

This specification is approved for use within U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a check valve used for draining fuel.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

AMSC N/A FSC 2910

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2.2 Government dicuments.

2.2.1 Specifications and standards. The following specifications and standards forma part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.

DEPARTMENT OF DEFENSE

MIL-T-5624 - Turbine Fuel, Aviation, Grades JP4 and JP5.
MIL-C-7024 - Calibrating Fluid, Aircraft Fuel System Components.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 191 11-5094.)

2.2.2 Government drawings The following Government drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation. .

DRAWINGS

DEPARTMENT OF DEFENSE

12302407 - Valve, Fuel-Drain.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren MI 48397-5000.)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the non-government documents that are curent on the date of the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI 21.4 - Sampling Procedures and TAbles for Inspections by Attributes.

(Copies of ANSI Standards maybe obtained from The American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for Valves. The materials shall be capable of meeting all of the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable (see 4.4.1).

3.2.1 Metallic materials. Metallic materials shall be either corrosion resistant or protected to prevent corrosion (see 4.4.1).

3.2.2 Compatibility. The valve shall be capable of functioning with, and shall not deteriorate from contact with, the following fluids (see 4.4. 1).

a. Turbine fuel per MIL-T-5624.
b. Calibration fluid per MIL-C-7024.
c. Diesel fuel oil per W-F-800.

3.3 Design and construction

3.3.1 Envelope. The envelope of the valve shall be as specified on Army Drawing 12302407.

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3.4 Performance.

3.4.1 Open-close pressure. When pressure is applied in the closed flow path, the valve shall close between 55 kilo Pascal (kPa) and 75 kPa (between 8 and 11 pounds per square inch gage (psig)). When the air pressure is decreased, the valve shall open between 75 and 55 kPa (between 11 and 8 psig) (see 4.4.4).

3,4.2 Proof pressure. The valve shall be capable of operating at 2068 kPa (300 psig) and shall be capable of withstanding a proof pressure of 3103 kpa (450 psig) with no signs of damage or external leakage (see 4.4.5).

3.4.3 Burst pressure. The valve shall be capable of withstanding a burst pressure of 5171 kpa (750 psig) without rupturing. The valve shall not have to be operational after meeting this requirement (see 4.4.6).

3.4.4 Leakage. Leakage of calibration fluid conforming to MIL-C-7024, type II, or an equivalent, at 4137 ± 138 kPa (600 ± 20 psig) through the closed flow path shall be not more than 0.5 cubic centimeter per minute (ccm). There shall be no external leakage (see 4.4.7).

3.4.5 Drainage. The valve shall drain calibration fluid conforming to MIL-C-7024, type II, or an equivalent, at a rate of not less than 3500 ccm with a pressure of 3.45 kPa (0.5 psig) (see 4.4.8).

3.4.6 Enviromental. The valve shall not be damaged and shall meet the requirements of 3.4.1 to 3.4.5 during exposure to the following environments (see 4.4.9):

a. Ambient temperature range of -51 to 371 degrees Celsius (“C) (-60 to 700 degrees Fahrenheit (“F)).
b. Fluid temperature range of -51 to 371°C (-60 to 700°F).

3.5 Identification and markings. The valve shall be permanently marked for identification in accordance with Army Drawing 12302407 (see 4.4.3).

3.6 Workmanship. Valves shall be processed in such a manner as to be uniform in quality and free of defects that will affect their life, serviceability, or appearance (see 4.4.3).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.2).
b. Quality conformance inspections (see 4.3).
   1. Acceptance tests (see 4.3.2).
   2. Control tests (see 4.3.3).

4.2 First article inspection. First article inspection shall be performed on a sample of valves produced under the production contract when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.3 (see table I) and the tests of table II.

TABLE I. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Valve, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
103	Identification marking, improper (see 3.5).	Visual
104	Faulty workmanship affecting performance (see 3.6).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE
202	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

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TABLE II. Classification of inspections.

Title	Requirement	Inspec-tion	First article	Quality conformance	
				Acceptance (100%)	Con-trol
Materials and construction	3.2 thru 3.2.2	4.4.1	X		
Defects (see 4.2 and table I)	3.3.1, 3.5, 3.6	4.4.3	X		
Open-close pressure	3.4.1	4.4.4	X	X	
Proof pressure	3.4.2	4.4.5	X	X	
Burst pressure	3.4.3	4.4.6	X		
Leakage	3.4.4	4.4.7	X		X
Drainage	3.4.5	4.4.8	X		X
Environmental	3.4.6	4.4.9	X		X

4.3 Quality conformance inspection.

4.3.1 Sampling. Samples from an inspection lot for conformance inspection shall be - selected in accordance with ANSI 21.4. Each sample selected shall be inspected as specified herein for the defects listed in table I. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found. First article test samples . shall not be shipped.

4.3.2 Acceptance tests (100percent). Each valve selected in accordance with 4.3.1 shall be subjected to the tests specified in table II. Failure of any test shall be cause for rejection.

4.3.3 Control tests. Control tests shall be conducted on one valve from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The valves shall be subjected to the trots specified in table H.

4.4 Methods of inspection.

4.4.1 Materials. Conformance to 3.2 thru 3.2.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 21 ± 10°C (70 ± 20°F)
b. Relative humidity: 50+ 30 percent

4.4.3 Defects. Conformance to 3.3.1, 3.5 and 3.6 shall be determined by examination for the defects listed in table 1. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.4.4 Open-close pressure test. To determine conformance to 3.4.1, submit the valve to increasing pressure and record when the valve poppet begins to close and when it fully closes. Then decrease the pressure, and record when the valve poppet begins to re-open and when it fully opens.

4.4.5 Proof pressure test. To determine conformance to 3.4.2, submit the valve to a pressure of 3103 kPa (450 psig) for 5 minutes, then examine it for damage or external leakage.

4.4.6 Burst pressure test. To determine conformance to 3.4.3, submit the valve to a pressure of 5171 kPa (750 psig), then examine it for signs of rupture.

4.4.7 Leakage test. To determine conformance to 3.4.4, subject the valve to a fluid pressure of 4137+ 138 kPa (600 ± 20 psig), measure any leakage and record. Then examine the valve for external leakage.

4.4.8 Drainage test. To determine conformance to 3.4.5, submit the valve to a fluid pressure of 3.45 kpa (0.5 psig), and measure and record the rate of flow.

4.4.9 Enviromental tests. To determine conformance to 3.4.6, submit the valve to the tests specified in 4.4.3 through 4.4.7 at the following environments:

a. Ambient and fluid temperature of 371 ‘C (700 ‘F).
b. Ambient and fluid temperatures of -51 “C (-60°F).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contractor order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s

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System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory.)

6.1 Inatended use. The valve covered by this specification is intended to be used for drainage from the combustor housing assembly.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of applicable drawings (see 2.2.2).
c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article is required (see 3. 1).
e. Inspection conditions" if other than as specified (see 4.4.2).
f. Selection of applicable packaging requirements (see 5.1).

6.3 Subject term (key word) listings.

D r a i n a g e
Draining
Fueling

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2910-0237)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-62625A(AT)	951208

3. DOCUMENT TITLE
VALVE, FUEL-DRAIN

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (810) 574-8745	(2) AUTOVON 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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